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Skill week-1

Creating Database, Table and Inserting Data Into Table.

SQL File 3*

Limit to 1000 rows

```sql
create DATABASE sweek1;

USE sweek1;

CREATE TABLE student_marks1 (
    roll_no INT PRIMARY KEY,
    name VARCHAR(50),
    subject VARCHAR(50),
    marks DECIMAL(5,2)
);

INSERT INTO student_marks1 (roll_no, name, subject, marks) VALUES
(1, 'Ravi', 'Math', 85.50),
(2, 'Sita', 'Math', 92.75),
(3, 'Anil', 'Math', 78.40),
(4, 'Priya', 'Math', 88.90),
(5, 'Vijay', 'Math', 80.25);
```

Output

Action Output

#	Time	Action	Message
1	10:58:37	create DATABASE sweek1	Error Code: 1007. Can't create database 'sweek1'; database exists
2	10:58:44	USE sweek1	0 row(s) affected
3	10:58:49	CREATE TABLE student_marks1 (roll_no INT PRIMARY KEY, name VARCHAR(50), subject VARCHAR(50), ...	0 row(s) affected
4	10:59:17	INSERT INTO student_marks (roll_no, name, subject, marks) VALUES (1, 'Ravi', 'Math', 85.50), (2, 'Sita', 'Math', 92.7...	Error Code: 1064. You have an error in your SQL syntax; check the manual that correspon
5	10:59:39	INSERT INTO student_marks (roll_no, name, subject, marks) VALUES (1, 'Ravi', 'Math', 85.50), (2, 'Sita', 'Math', 92.7...	Error Code: 1146. Table 'sweek1.student_marks' doesn't exist
6	10:59:52	INSERT INTO student_marks1 (roll_no, name, subject, marks) VALUES (1, 'Ravi', 'Math', 85.50), (2, 'Sita', 'Math', 92....	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0

```sql
SELECT * FROM student_marks1;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
3	Anil	Math	78.40
4	Priya	Math	88.90
5	Vijay	Math	80.25
NULL	NULL	NULL	NULL

dent_marks1 1

Output

Action Output

#	Time	Action
1	11:20:04	use sweek1
2	11:20:21	SELECT * FROM student_marks1 LIMIT 0, 1000

```sql
SELECT * FROM student_marks1;

SELECT COUNT(*) AS total_students FROM student_marks1;
SELECT SUM(marks) FROM student_marks1;
SELECT AVG(marks) FROM student_marks1;
SELECT MAX(marks) FROM student_marks1;
SELECT MIN(marks) FROM student_marks1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

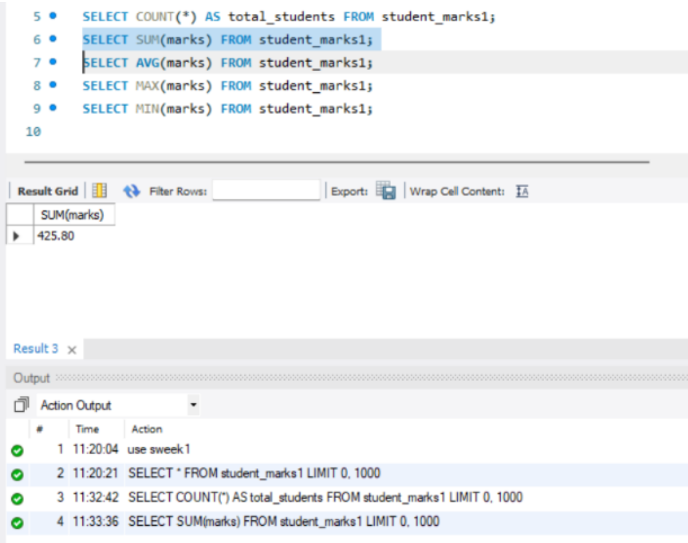
total_students
5

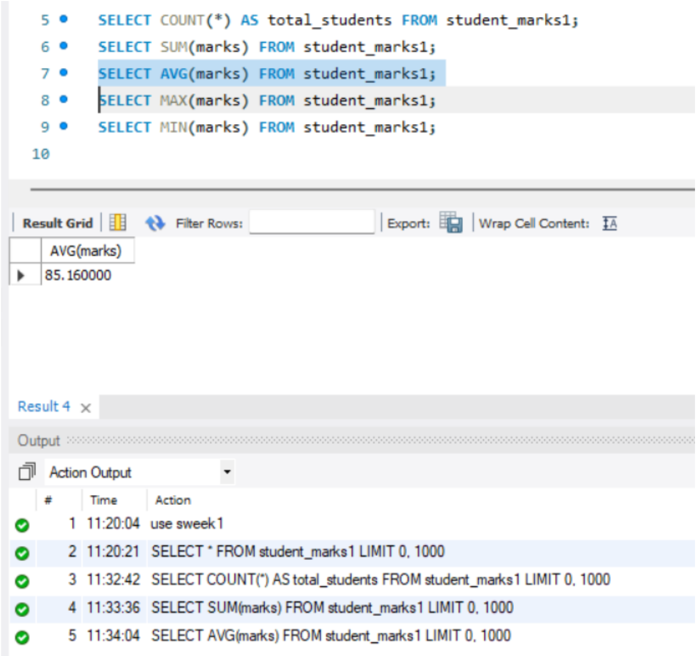
Result 2

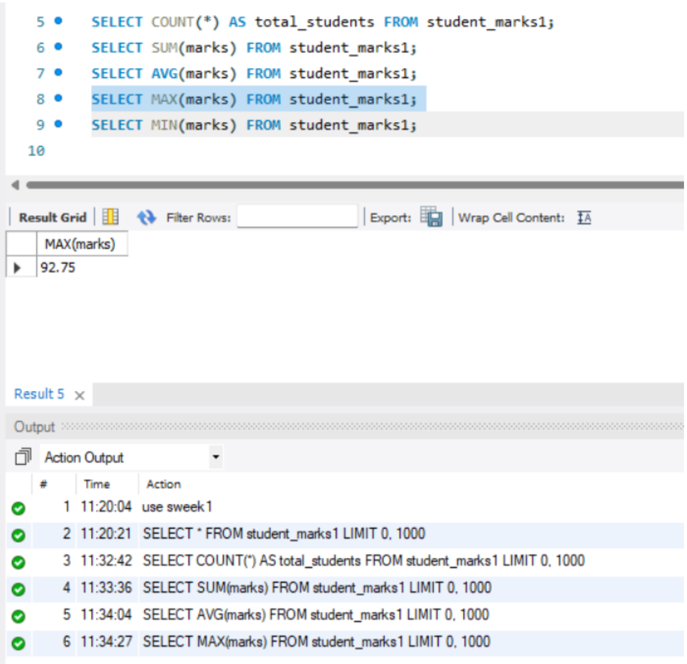
Output

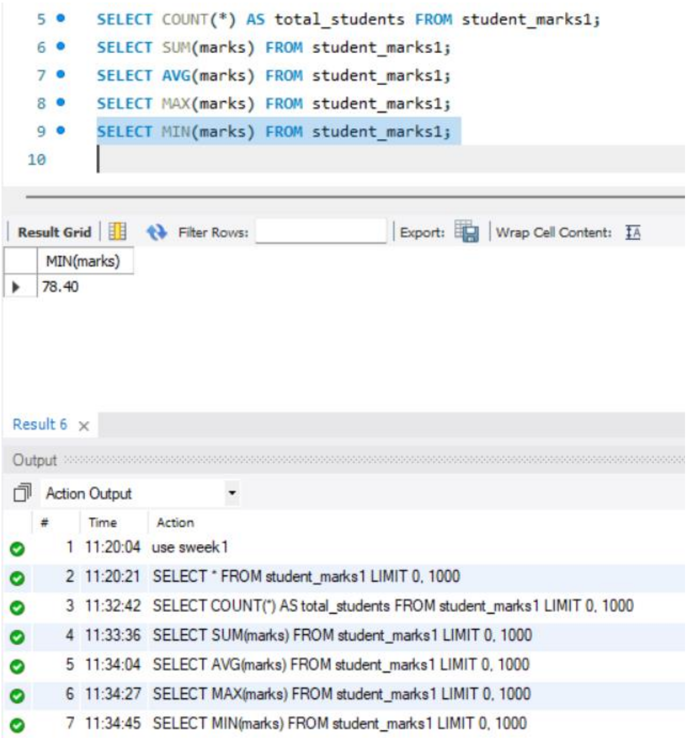
Action Output

#	Time	Action
1	11:20:04	use sweek1
2	11:20:21	SELECT * FROM student_marks1 LIMIT 0, 1000
3	11:32:42	SELECT COUNT(*) AS total_students FROM student_marks1 LIMIT 0, 1000

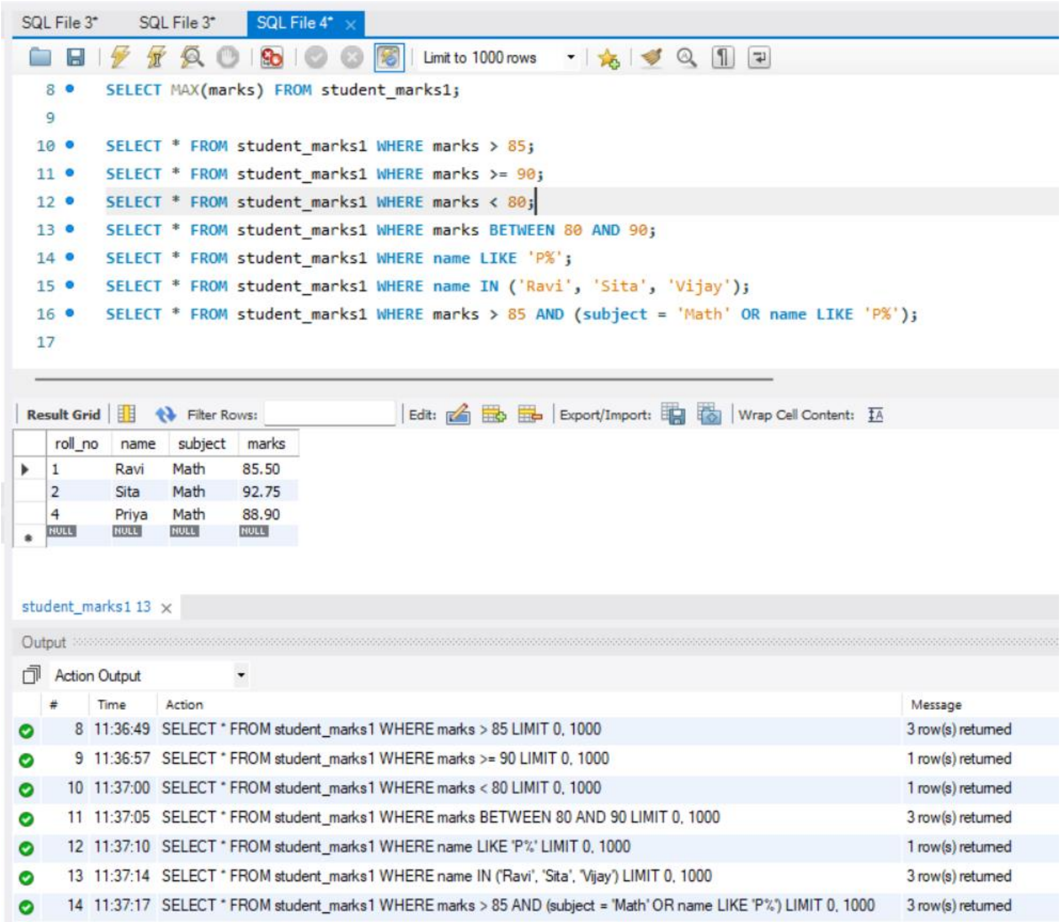








Filtering with WHERE



SQL File 3* | SQL File 3* | SQL File 4*

Limit to 1000 rows

```sql
SELECT MAX(marks) FROM student_marks1;

SELECT * FROM student_marks1 WHERE marks > 85;
SELECT * FROM student_marks1 WHERE marks >= 90;
SELECT * FROM student_marks1 WHERE marks < 80;
SELECT * FROM student_marks1 WHERE marks BETWEEN 80 AND 90;
SELECT * FROM student_marks1 WHERE name LIKE 'P%';
SELECT * FROM student_marks1 WHERE name IN ('Ravi', 'Sita', 'Vijay');
SELECT * FROM student_marks1 WHERE marks > 85 AND (subject = 'Math' OR name LIKE 'P%');
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
4	Priya	Math	88.90
NULL	NULL	NULL	NULL

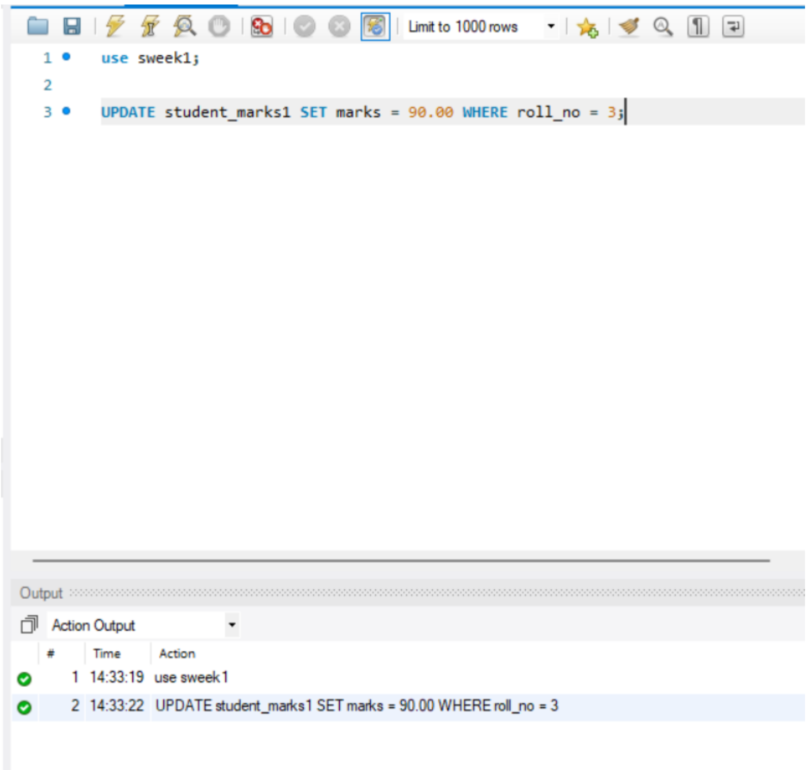
student_marks1 13 ×

Output

Action Output

#	Time	Action	Message
8	11:36:49	SELECT * FROM student_marks1 WHERE marks > 85 LIMIT 0, 1000	3 row(s) returned
9	11:36:57	SELECT * FROM student_marks1 WHERE marks >= 90 LIMIT 0, 1000	1 row(s) returned
10	11:37:00	SELECT * FROM student_marks1 WHERE marks < 80 LIMIT 0, 1000	1 row(s) returned
11	11:37:05	SELECT * FROM student_marks1 WHERE marks BETWEEN 80 AND 90 LIMIT 0, 1000	3 row(s) returned
12	11:37:10	SELECT * FROM student_marks1 WHERE name LIKE 'P%' LIMIT 0, 1000	1 row(s) returned
13	11:37:14	SELECT * FROM student_marks1 WHERE name IN ('Ravi', 'Sita', 'Vijay') LIMIT 0, 1000	3 row(s) returned
14	11:37:17	SELECT * FROM student_marks1 WHERE marks > 85 AND (subject = 'Math' OR name LIKE 'P%') LIMIT 0, 1000	3 row(s) returned

Updating Queries



Limit to 1000 rows

```sql
use sweek1;

UPDATE student_marks1 SET marks = 90.00 WHERE roll_no = 3;
```

Output

Action Output

#	Time	Action
1	14:33:19	use sweek1
2	14:33:22	UPDATE student_marks1 SET marks = 90.00 WHERE roll_no = 3

Deleting Queries

```sql
use sweek1;

UPDATE student_marks1 SET marks = 90.00 WHERE roll_no = 3;

DELETE FROM student_marks1 WHERE roll_no = 5;
DELETE FROM student_marks1 WHERE marks < 75;
```

Output

Action Output

#	Time	Action
1	14:33:19	use sweek1
2	14:33:22	UPDATE student_marks1 SET marks = 90.00 WHERE roll_no = 3
3	14:41:35	DELETE FROM student_marks WHERE roll_no = 5
4	14:41:51	DELETE FROM student_marks1 WHERE roll_no = 5
5	14:41:54	DELETE FROM student_marks1 WHERE marks < 75

Sorting

```sql
SELECT * FROM student_marks1 ORDER BY marks ASC;
SELECT * FROM student_marks1 ORDER BY marks DESC;
SELECT * FROM student_marks1 ORDER BY name ASC;
SELECT * FROM student_marks1 ORDER BY name DESC;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

roll_no	name	subject	marks
1	Ravi	Math	85.50
4	Priya	Math	88.90
3	Anil	Math	90.00
2	Sita	Math	92.75
NULL	NULL	NULL	NULL

dent_marks1 1 ×

Output

Action Output

#	Time	Action
1	14:33:19	use sweek1
2	14:33:22	UPDATE student_marks1 SET marks = 90.00 WHERE roll_no = 3
3	14:41:35	DELETE FROM student_marks WHERE roll_no = 5
4	14:41:51	DELETE FROM student_marks1 WHERE roll_no = 5
5	14:41:54	DELETE FROM student_marks1 WHERE marks < 75
6	14:44:25	SELECT * FROM student_marks1 ORDER BY marks ASC LIMIT 0, 1000

```sql
SELECT * FROM student_marks1 ORDER BY marks ASC;
SELECT * FROM student_marks1 ORDER BY marks DESC;
SELECT * FROM student_marks1 ORDER BY name ASC;
SELECT * FROM student_marks1 ORDER BY name DESC;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

roll_no	name	subject	marks
2	Sita	Math	92.75
3	Anil	Math	90.00
4	Priya	Math	88.90
1	Ravi	Math	85.50
NULL	NULL	NULL	NULL

student_marks1 2 ×

Output

Action Output

#	Time	Action
1	14:33:19	use sweek1
2	14:33:22	UPDATE student_marks1 SET marks = 90.00 WHERE roll_no = 3
3	14:41:35	DELETE FROM student_marks WHERE roll_no = 5
4	14:41:51	DELETE FROM student_marks1 WHERE roll_no = 5
5	14:41:54	DELETE FROM student_marks1 WHERE marks < 75
6	14:44:25	SELECT * FROM student_marks1 ORDER BY marks ASC LIMIT 0, 1000
7	14:45:15	SELECT * FROM student_marks1 ORDER BY marks DESC LIMIT 0, 1000

```sql
SELECT * FROM student_marks1 ORDER BY marks ASC;
SELECT * FROM student_marks1 ORDER BY marks DESC;
SELECT * FROM student_marks1 ORDER BY name ASC;
SELECT * FROM student_marks1 ORDER BY name DESC;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

roll_no	name	subject	marks
2	Sita	Math	92.75
1	Ravi	Math	85.50
4	Priya	Math	88.90
3	Anil	Math	90.00
NULL	NULL	NULL	NULL

student_marks1 4 ×

Output

Action Output

#	Time	Action
3	14:41:35	DELETE FROM student_marks WHERE roll_no = 5
4	14:41:51	DELETE FROM student_marks1 WHERE roll_no = 5
5	14:41:54	DELETE FROM student_marks1 WHERE marks < 75
6	14:44:25	SELECT * FROM student_marks1 ORDER BY marks ASC LIMIT 0, 1000
7	14:45:15	SELECT * FROM student_marks1 ORDER BY marks DESC LIMIT 0, 1000
8	14:47:45	SELECT * FROM student_marks1 ORDER BY name ASC LIMIT 0, 1000
9	14:47:52	SELECT * FROM student_marks1 ORDER BY name DESC LIMIT 0, 1000

Group By

```sql
SELECT subject, SUM(marks) AS total_marks FROM student_marks1 GROUP BY subject;
SELECT subject, AVG(marks) AS avg_marks FROM student_marks1 GROUP BY subject;
SELECT subject, COUNT(*) AS num_students FROM student_marks1 GROUP BY subject;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

subject	num_students
Math	4

Result 7 ×

Output

Action Output

#	Time	Action
6	14:44:25	SELECT * FROM student_marks1 ORDER BY marks ASC LIMIT 0, 1000
7	14:45:15	SELECT * FROM student_marks1 ORDER BY marks DESC LIMIT 0, 1000
8	14:47:45	SELECT * FROM student_marks1 ORDER BY name ASC LIMIT 0, 1000
9	14:47:52	SELECT * FROM student_marks1 ORDER BY name DESC LIMIT 0, 1000
10	14:48:56	SELECT subject, SUM(marks) AS total_marks FROM student_marks1 GROUP BY subject LIMIT 0, 1000
11	14:49:01	SELECT subject, AVG(marks) AS avg_marks FROM student_marks1 GROUP BY subject LIMIT 0, 1000
12	14:49:05	SELECT subject, COUNT(*) AS num_students FROM student_marks1 GROUP BY subject LIMIT 0, 1000

Having

```sql
SELECT subject, AVG(marks) AS avg_marks
FROM student_marks1
GROUP BY subject
HAVING AVG(marks) > 90;

SELECT subject, COUNT(*) AS num_students
FROM student_marks1
GROUP BY subject
HAVING COUNT(*) > 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	subject	avg_marks

Result 8 ×

Output

Action Output

	#	Time	Action
✓	8	14:47:45	SELECT * FROM student_marks1 ORDER BY name ASC LIMIT 0, 1000
✓	9	14:47:52	SELECT * FROM student_marks1 ORDER BY name DESC LIMIT 0, 1000
✓	10	14:48:56	SELECT subject, SUM(marks) AS total_marks FROM student_marks1 GROUP BY subject LIMIT 0, 1000
✓	11	14:49:01	SELECT subject, AVG(marks) AS avg_marks FROM student_marks1 GROUP BY subject LIMIT 0, 1000
✓	12	14:49:05	SELECT subject, COUNT(*) AS num_students FROM student_marks1 GROUP BY subject LIMIT 0, 1000
✗	13	14:50:08	SELECT subject, AVG(marks) AS avg_marks FROM student_marks GROUP BY subject HAVING AVG(marks) > 90...
✓	14	14:50:22	SELECT subject, AVG(marks) AS avg_marks FROM student_marks1 GROUP BY subject HAVING AVG(marks) > 9...

```sql
SELECT subject, AVG(marks) AS avg_marks
FROM student_marks1
GROUP BY subject
HAVING AVG(marks) > 90;

SELECT subject, COUNT(*) AS num_students
FROM student_marks1
GROUP BY subject
HAVING COUNT(*) > 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	subject	num_students
▶	Math	4

Result 9 ×

Output

Action Output

	#	Time	Action
✓	9	14:47:52	SELECT * FROM student_marks1 ORDER BY name DESC LIMIT 0, 1000
✓	10	14:48:56	SELECT subject, SUM(marks) AS total_marks FROM student_marks1 GROUP BY subject LIMIT 0, 1000
✓	11	14:49:01	SELECT subject, AVG(marks) AS avg_marks FROM student_marks1 GROUP BY subject LIMIT 0, 1000
✓	12	14:49:05	SELECT subject, COUNT(*) AS num_students FROM student_marks1 GROUP BY subject LIMIT 0, 1000
✗	13	14:50:08	SELECT subject, AVG(marks) AS avg_marks FROM student_marks GROUP BY subject HAVING AVG(marks) > 90...
✓	14	14:50:22	SELECT subject, AVG(marks) AS avg_marks FROM student_marks1 GROUP BY subject HAVING AVG(marks) > 9...
✓	15	14:50:50	SELECT subject, COUNT(*) AS num_students FROM student_marks1 GROUP BY subject HAVING COUNT(*) > 1...